Experiment 2: at-max message

Follows the ship decision in DECISIONS.pdf. Population: members already at their maximum (max eligible limit minus current limit under $100), about 23% of members.

> **Hypothesis:** members at their max can't get an increase, and the shipped app now shows them nothing. A clear, honest explanation that they're at their maximum, with no request, no credit check and no decline, will retain them **better than silence**. It could also read as a "no" and hurt, which is why it's tested rather than shipped.

Background

In the first experiment, members at their max who could request an increase retained 18 to 26 pp worse than control at every window. Their maximum is fixed at signup, so any request was above it and was declined, and they were shown a decline reason that wasn't the real cause. Most who closed did so within a week of the decline. The shipped version hides the Increase button from these members, which returns them to control retention. That shipped state is arm A here. Their maximum stays fixed, so neither arm can offer or promise an increase.

Arms

	A: No button (shipped)	B: At-max message
Home	Credit limit and Spent only; no Increase button	A card above the account details: "You're at your maximum limit. Your maximum was set when you joined, and your $X limit has reached it, so an increase isn't available on this account."
Reassurance	None	"Nothing here checks your credit, and there's nothing you need to do."
Useful next step	None	Shows Available to spend (limit minus spent and pending holds)
Request, credit check, decline	None	None

Members with headroom are in experiment 1 instead. Members who already requested see neither arm.

Design and metrics

Assignment	Randomized 50/50 at signup among members with under $100 of headroom; headroom is known at signup and fixed, so eligibility can't be influenced by the arm
Primary metric	30-day retention (account open 30 days after joining), B minus A, with 95% CI. 60-day retention read as follow-up
Secondary	7- and 14-day retention, since the original harm was early; how often members view the card
Guardrails	Support contacts about limits; any early-retention drop in B
Sample size	Arm A's 30-day retention should match control for these members, about 87.9%. Detecting a 1 pp difference at 80% power and 5% significance needs about 16,700 members per arm. At the first experiment's signup rate, that's roughly 10 days to two weeks of enrollment, then 30 (and 60) days to read
Ship rule	Ship B if 30-day retention is higher with a CI excluding zero and early retention is no worse. If B is lower, keep A. If there's no difference, keep A, the simpler experience

Try it in the app

The React Native app includes both arms. Start the mock API, then the web app, in two terminals:

```
cd mock-api && npm start
cd react_native_app && npm install && npm run web
```

- On Home, set **PERSONA** to **insufficient-history**, the only persona at its max.
- Use **ARM** to switch arms. Its labels follow the persona's experiment: for this persona they read **A: No button** and **B: At-max message**; for personas with headroom they read A: Request and B: Pre-approved (experiment 1). The arm control is for demos; in production the arm comes from assignment.

Persona	Limit / max eligible	Arm A	Arm B
insufficient-history	$1,500 / $1,500	No Increase button; Credit limit $1,500.00 and Spent $310.00	"You're at your maximum limit" card naming the $1,500 limit, plus Available to spend $1,178.00 ($1,500 minus $310 spent and $12 pending)
approved, utilization-too-high, new-credit-lines	Room of $500 to $2,500	Experiment 1, arm A	Experiment 1, arm B

Implementation notes

- All changes are in `react_native_app/App.tsx`. No API changes: available to spend is derived as `limit_cents - balance_cents - pending_holds_cents` from GET /v1/credit-line.
- A member is at their max when they haven't requested and their room is under the API's $100 minimum increase. The message is in `AtMaxCard`.
- One A/B arm setting drives both experiments; which experiment applies is decided by the member's headroom.
- No copy implies a future increase, since maximum eligible limits are fixed.